

NAGA VENKATA BHARATH LANKA

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EDUCATION

MASTERS in DATA SCIENCE • UNIVERSITY OF NEW HAVEN • NEW HAVEN • 2025 • 3.5

PROFESSIONAL EXPERIENCE

DATA ANALYST- TECH MAHINDRA

December 2021 – July 2023, HYDERABAD

- Extracted, cleaned, and analyzed large-scale datasets (10M+ records) using SQL and Python, improving data accuracy and reducing reporting errors by 25%.
- Built automated ETL pipelines in Python, reducing manual reporting time by 40% and ensuring timely availability of insights for business teams.
- Designed and deployed interactive dashboards in Power BI to track KPIs, enabling executives to make faster, data-driven decisions and increasing operational efficiency by 15%.
- Conducted exploratory data analysis (EDA) and applied statistical modeling (regression, clustering, hypothesis testing) to uncover patterns and trends, supporting strategic workforce planning.
- Collaborated with cross-functional teams to implement predictive models (customer churn, sales forecasting), achieving up to 85% accuracy and reducing churn by 10%.
- Optimized queries and database structures, reducing data retrieval time by 30%.

SKILLS

- Programming & Databases: Python, SQL
- Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Model Optimization
- Deep Learning: Neural Networks (CNN, RNN, LSTM), Transfer Learning
- Natural Language Processing (NLP): Tokenization, Sentiment Analysis, Text Summarization
- Analytics & BI: Microsoft Excel, Power BI, Power Query
- Data Management: Data Modeling, EDA, Data Visualization, ETL, Data Pipelines
- Frameworks & Libraries: Pandas, NumPy, TensorFlow, PyTorch, Scikit-learn, GitHub
- Cloud Platforms: AWS, Azure, GCP
- Big Data Technologies: Spark, Hadoop

PROJECTS

AI PROJECT- Implementation Of Baroque Chess | September 2023 - December 2023

- Designed and implemented an AI agent for Baroque Chess, a chess variant with unique rules and pieces, using the Minimax algorithm with Alpha-Beta pruning.
- Built an evaluation function to assess board positions, optimizing decision-making efficiency by 20% compared to baseline approaches.
- Demonstrated application of search algorithms, heuristic evaluation, and game theory in complex decision-making environments.

Technologies & Skills: Python, AI search algorithms, Alpha-Beta pruning, heuristic evaluation, game theory.

Neural Network-based Language Model for Next Token Prediction | Fall 2024

- Developed NN, RNN, and LSTM models for next-token prediction in English and Estonian using HuggingFace datasets.
- Performed text preprocessing including tokenization, lowercasing, and cleaning to prepare data for model training.
- Trained and optimized models using cross-entropy loss and Adam optimizer; monitored learning curves for performance.
- LSTM model achieved highest accuracy, generating coherent multilingual text sequences.

Technologies & Skills: Python, PyTorch, HuggingFace, NLP preprocessing, sequence modeling, multilingual text generation.

Real-Time Object Detection Using YOLOv8s and OpenCV • September 2024 - December 2024

- Implemented a YOLOv8-based deep learning model for real-time object detection using a custom bathroom dataset.
- Leveraged pre-trained COCO weights and fine-tuned the model, achieving 95% detection accuracy across multiple object categories.
- Integrated with OpenCV to enable live webcam detection, reducing inference latency by 30% while maintaining precision.
- Analyzed the YOLOv8 architecture (backbone, neck, detection heads) and optimized feature extraction, improving segmentation and object recognition reliability.
- Demonstrated applicability to smart home automation and real-time monitoring systems.

CERTIFICATIONS

DATA SCIENCE FOUNDATION- DATA ENGINEER

LINKEDIN- LEARNINGS • 2024

SQL Essential Training (2019)

LINKEDIN- LEARNINGS • 2024
